

# Applications of Queues

Q. Discuss the significance of Queues in computer science. Elaborate on various real-world and system-level applications where the Queue data structure is essential.

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**Definition to Remember**

A **Queue** is a linear data structure that follows the First In First Out (FIFO) principle. It allows elements to be added (enqueue) at one end and removed (dequeue) from the other end, ensuring that elements are processed in the order they were added.

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## 1. System-Level Applications (Operating Systems & Networks)

| Application | Description |

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## 2. Real-World & Algorithmic Applications

| Application | Description |

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## 3. Print Spooler Example Diagram

text

[User 1

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**Must-Write Points (for 10 marks)**

1. Queue

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***\*\*Quick Recall\*\****

***`Queue  
Algorith***